CS 225

#4: The Rule of Three

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Three ways to return a variable:

joinSpheres-returnByValue.cpp

```
Sphere joinSpheres(const Sphere &s1, const Sphere &s2) {
  double totalVolume = s1.getVolume() + s2.getVolume();

  double newRadius = std::pow(
    (3.0 * totalVolume) / (4.0 * 3.141592654),
    1.0/3.0
  );

  return Sphere(newRadius);
}
```

joinSpheres-returnByPointer.cpp

```
Sphere *joinSpheres(const Sphere &s1, const Sphere &s2) {
  double totalVolume = s1.getVolume() + s2.getVolume();

  double newRadius = std::pow(
    (3.0 * totalVolume) / (4.0 * 3.141592654),
    1.0/3.0
  );

  return new Sphere(newRadius);
}
```

joinSpheres-returnByReference.cpp

```
Sphere & joinSpheres(const Sphere &s1, const Sphere &s2) {
  double totalVolume = s1.getVolume() + s2.getVolume();

  double newRadius = std::pow(
    (3.0 * totalVolume) / (4.0 * 3.141592654),
    1.0/3.0
  );
  //
  // BAD PRATICE: ____________________________________
  //
  Sphere *result = new Sphere(newRadius);
  return *result; // Memory must be delete'd later
}
```

```
  //
  // ERROR: __________________________________________
  //
  Sphere result(newRadius);
  return result;
}
```

Return by reference takeaway:

Adding our Sphere class:

What should happen when we add two `Sphere` classes?

addSpheres.cpp

```
#include "sphere.h"

int main() {
  cs225::Sphere s1(3), s2(4);
  cs225::Sphere s3 = s1 + s2;
  return 0;
}
```

Overloading Operators

C++ allows custom behaviors to be defined on over 20 operators:

Arithmetic	+ - * / % ++ --
Bitwise	& | ^ ~ << >>
Assignment	=
Comparison	== != > < >= <=
Logical	! && ||
Other	[] () ->

General Syntax:

Adding overloaded operators to Sphere:

sphere.h

```
#ifndef SPHERE_H
#define SPHERE_H

class Sphere {
  public:
    // ...







    // ...
  private:
    double r_;
};

#endif
```

sphere.cpp

```
/* ... */















/* ... */
```

One Very Special Operator: ______________________

1. Similar to the Copy Constructor:

2. Different than the Copy Constructor:

sphere.h

```
#ifndef SPHERE_H
#define SPHERE_H

class Sphere {
  public:
    // ...





    // ...
  private:
    double r_;
};

#endif
```

sphere.cpp

```
/* ... */















/* ... */
```

Big Idea: Destructor

sphere.h

```
#ifndef SPHERE_H
#define SPHERE_H

class Sphere {
  public:
    // ...


    // ...
```

sphere.cpp

```
/* ... */







/* ... */
```

The "Rule of Three":

1.

2.

3.

Why are the big three so important?

Example:

sphere.h

```
#ifndef SPHERE_H
#define SPHERE_H

class Sphere {
  public:
    // ...


    // ...
```

sphere.cpp

```
/* ... */







/* ... */
```

CS 225 – Things To Be Doing:

1. How was your quiz?
2. MP1 due on Monday
3. MP2 released after MP1 is due *(start early for +EC!)*
4. Happy Winter break!